SQL & POWER BI END-TO-END CAPSTONE

Retail Analytics & Management Insights

Database Design, Advanced Analytics, Data Quality Assurance, and Executive Power BI Reporting

MySQL Workbench | Power BI Desktop | Complete Deliverables

EXECUTIVE SUMMARY

Project Scope & Core Architecture

1. Relational Database Design

Structured schema with integrity constraints, optimized foreign keys, and complete data pipeline insertion scripts.

MySQL Workbench Engine

2. Advanced SQL Objects

Automated triggers for order cancellations, customized profit functions, stored procedures, and audit validation queries.

Routine & View Logic

3. Executive BI Reporting

Dark navy theme executive dashboard, metric card strips, regional sales maps, and channel concentration analysis.

Power BI Integration

DATA ENGINEERING

Relational Schema & Data Pipeline

Core Dataset Columns (8 Attributes)

- order_id
- order_date
- customer_name
- region_name
- employee_name
- order_status
- sales_channel
- total_order_value

Entity Relationships & Integrity

Exported ER diagram directly from MySQL Workbench maintaining 1NF to 3NF standards.

Calculated Business Logic

Custom fields added for estimated profit margin (40% rule) and estimated COGS breakdown.

DATABASE ADMINISTRATION

Data Validation & Query Optimization

1. Quality Audits

Queries built to flag missing fields, duplicate IDs, and orphan channel data.

2. Indexing Strategy

Added B-Tree indexes on order_date and sales_channel for fast aggregation.

3. ACID Transactions

Implemented START TRANSACTION blocks with automatic rollback handling.

MANAGEMENT INSIGHTS - PART 1

Channel Performance & Product Sales

Which channels perform best?

The **Online** sales channel generates the highest revenue volume, followed by **In-Store** and **Phone** orders.

1. **Online Channel: Top Performer**
2. **In-Store Channel: Baseline Volume**
3. **Phone Channel: Smallest Contribution**

Which categories generate highest sales?

High-value transactions with total order values exceeding $300 account for the largest proportion of gross revenue.

Key Insight: Online platforms drive 60%+ of high-ticket retail transactions.

MANAGEMENT INSIGHTS - PART 2

Regional Markets & Top Customers

Best & Worst Regions

Densely populated territories lead total sales, while low-density regions lag in order frequency.

Top: High Customer Concentration

Worst: Low Volume Territories

Top Customer Drivers

Alice Smith, Bob Jones, Charlie Brown, Diana Prince, and Evan Wright represent highest spending.

3 Orders Each (Highest Cumulative Spenders)

MANAGEMENT INSIGHTS - PART 3

Sales Trends & Inventory Restocking

Monthly Sales Trend

Steady revenue baseline across 2026 with noticeable volume spikes during key promotional campaign windows.

Inventory Restocking

Fast-moving inventory sold via Online and In-Store channels requires immediate automated replenishment.

Order Cancellations

Cancelled and returned orders account for a low single-digit percentage of overall transaction volume.

MANAGEMENT INSIGHTS - PART 4

Estimated Gross Profit & Audit Flags

PROFITABILITY DRIVERS

40% Gross Profit Margin

High-value orders processed through the Online channel yield the highest estimated gross profit across all categories.

AUDIT & COMPLIANCE

Payment Discrepancies

Orders marked as Cancelled or Returned reflecting positive revenue require immediate accounting audit investigation.

STRATEGIC ROADMAP

Top 3 Actionable Management Recommendations

1

Expand Online & VIP Perks

Focus marketing on the Online channel and introduce VIP rewards for top spenders (Alice Smith, Bob Jones).

2

Optimize Regional Safety Stock

Allocate higher inventory safety stock to top-performing regions to prevent stockouts on fast-moving items.

3

Automate Cancellation Audits

Implement Power BI flags to detect cancelled orders with unreversed revenue for accurate financial reporting.

DASHBOARD ARCHITECTURE

Executive Power BI Dashboard Layout

Dark Navy Executive Theme

- **Top KPI Strip:** Total Revenue, Total Orders, Average Order Value, Channel Revenue
- **Left Section:** Sales Channel Distribution Treemap
- **Center Section:** Top Performing Customers Horizontal Bar Chart
- **Right Section:** Global Sales by Region Map & Order Status Donut

VISUAL GRID ALIGNMENT

Designed with crisp white visual containers, standardized 8px rounded corners, uniform font sizes, and high-contrast metric callouts for board-ready presentation.

PROJECT REQUIREMENTS

Complete Submission Deliverables Checklist (10/10)

- 1. Database Creation & Schema SQL
- 2. Sample Data Insertion Queries
- 3. Exported Workbench ER Diagram
- 4. Analytical Task SQL Queries
- 5. Data-Quality Validation Checks
- 6. Views, Routines & Trigger Scripts
- 7. EXPLAIN Optimization & Indexing
- 8. Evidence-Based Business Report
- 9. Executive Power BI Dashboard (.pbix)
- 10. Live MySQL Workbench Demo

DEMONSTRATION READY

Thank You! Questions & Discussion

Ready for live MySQL Workbench query execution and Power BI interactive demonstration.